

Database Design and Development

Week 02
Database Environment

Database Models

- A set of concepts to describe the structure of a database and certain(சில) constraints that the database should obey.
 - Many forms of data models
 - Makes data abstraction possible
 - Hides details of physical data storage from the user

Structure of a database

- Data Model comprises:
 - a structural part;
 - a manipulative part;
 - possibly a set of integrity rules.
- Most commercial database systems are based on relational data model

Categories of data models

- *Data models can be categorized according to the types of concepts they use to describe the database structure :*
 - Conceptual (high-level, semantic) data models
 - Physical (low-level, internal) data models
 - Implementation (representational) data models

Conceptual (high-level, semantic) data models

- Provide concepts that are close to the way many users *perceive* data.
- Conceptual models include
 - Entity-relationship database model (ERDBM)
 - Object-oriented model (OODBM)

Conceptual data models, Cont'

- Conceptual data models use concepts such as entities, attributes, and relationships.
- Relationships in Conceptual Models
 - One-to-one (1:1)
 - One-to-many (1:M)
 - Many-to-many (M:N)

Physical (low-level, internal) data models

- Provide concepts that describe details of how data is stored in the computer.
 - Record formats, record orderings and access paths
 - Target computer specialists and not typical end users

Implementation (representational) data models

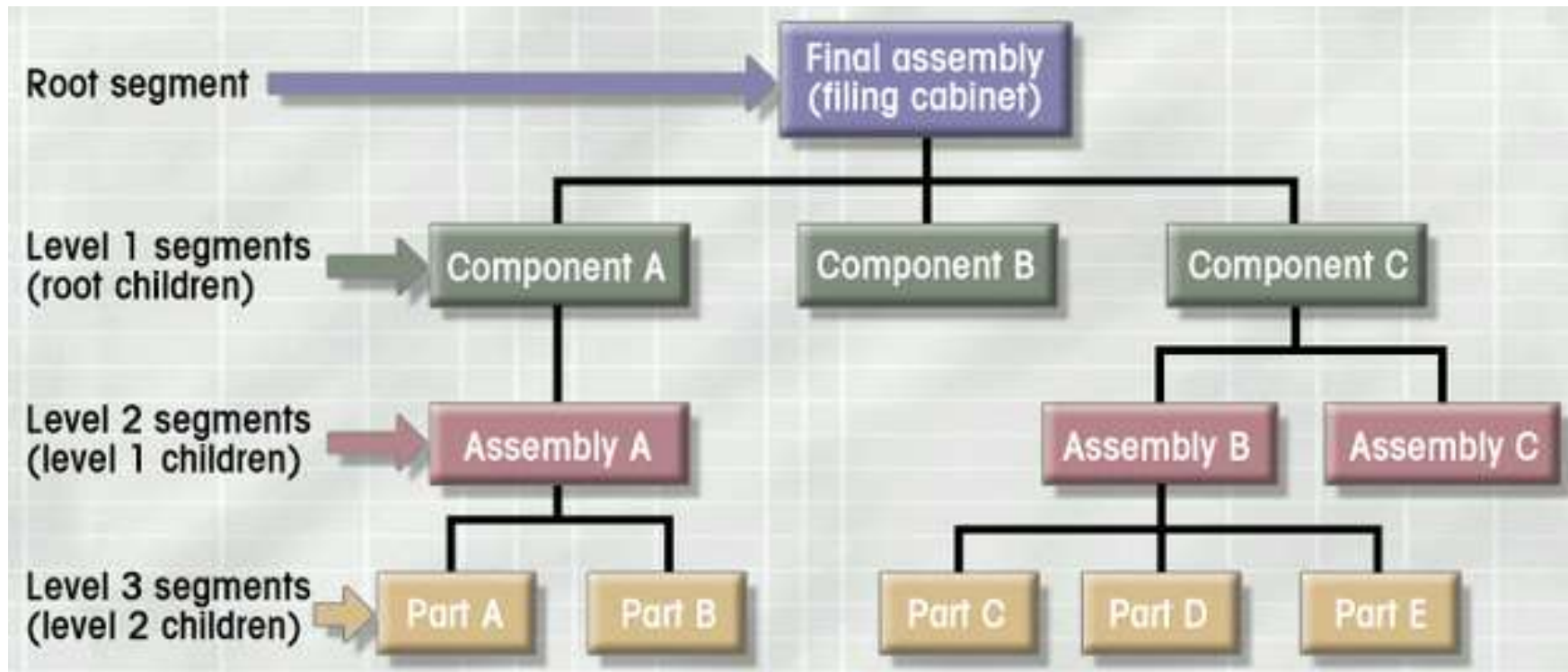
- Emphasizes on how the data is represented in the database or **how the data structures are implemented to represent what is modeled:**
- Implementation models include
 - **Hierarchical database model (HDBM)**
 - **Network database model (NDBM)**
 - **Relational database model (RDBM)**
 - **Object-oriented database model (ODBM)**

Hierarchical Database Model (HDBM)

- Logically represented by an upside down tree
 - Each parent can have many children (segment linkage)
 - Each child has only one parent

Hierarchical Database Model

- Logically represented by an upside down tree
- 1:M relationship



Hierarchical Database Model

- Hierarchical path (beginning from left)
- Left-list hierarchical path, or preorder traversal, or hierarchical sequence

**Final assembly->Component A->Assembly A-> -> Part A
->Part B -> Component B -> Component C -Assembly B
-> Part C ->Part D**

- Re-list sequence, if the segment is frequently accessed
- Bank systems commonly use HD model

Hierarchical Database Model

- Bank systems commonly use the HDBM
 - customer account can be subject to many transactions (1:M relationship)
 - Relationship is fixed (debiting and crediting)
 - Frequently access large amount of transactions

Hierarchical Database Model

- Advantages

- Conceptual simplicity: relationship between layers is logically simple; design process is simple
- Database security: enforced uniformly through the system
- Data integrity
- Data independence
- Efficiency in 1:M relationships and when uses require large numbers of transactions
- Dominant in 1970s , when we used mainframe system with large databases

Hierarchical Database Model

- Disadvantages

- Complex implementation: physical data storage characteristics; database design is complicated
- Difficult to manage and lack of standards
- Lacks structural independence
- Applications programming and use complexity (pointer based)
- Implementation limitations, i.e. especially it only handle 1:M type of model
 - large amount

Network Database Model (NDBM)

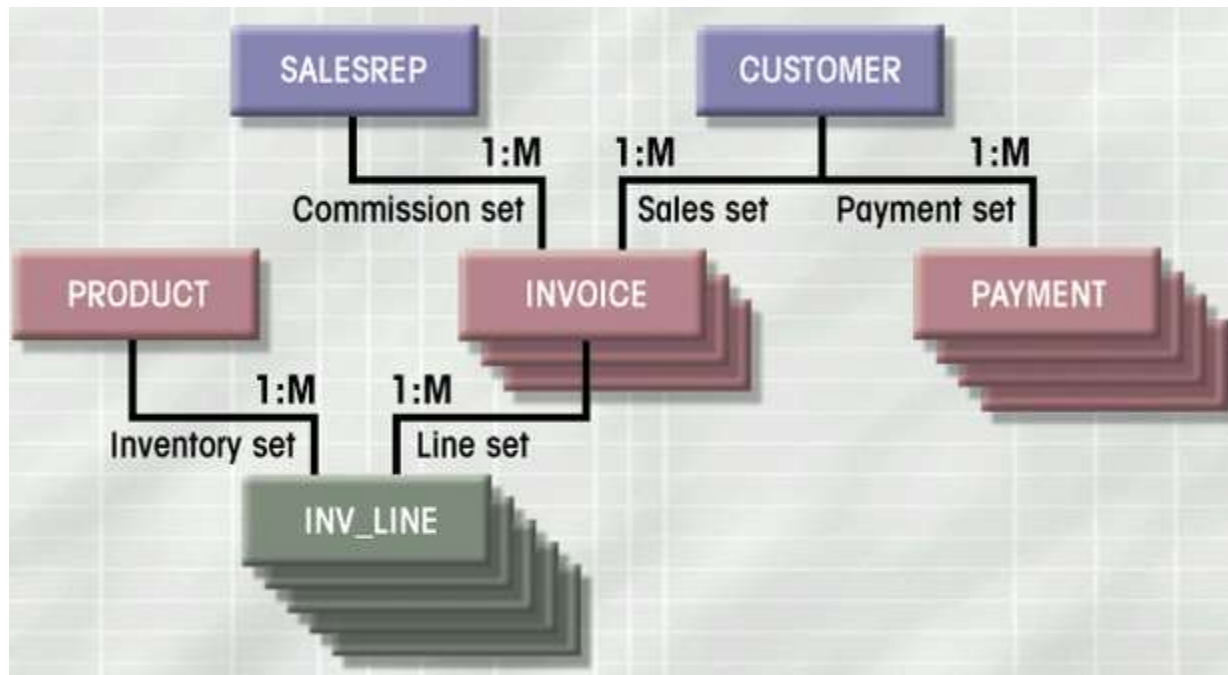
- Each record can have multiple parents
 - Called by Database Task Group (DBTG) to define standards
 - Three crucial database components
 - **Network schema**: conceptual organization of the entire database
 - **Subschema**: portion of database as information for application programs
 - **Database management language**: defining data characteristics and data structure
 - **Schema** Data definition language (**DDL**): define schema components
 - **Subschema** Data definition language(**ddl**)
 - **Data manipulating language**: manipulate data content

Network Database Model

- Each record can have multiple parents
- Introduce set to describe relationship
- Each set has **owner** record and **member** record,
parallel to parent and child in HDM
 - Member may have several owners
 - One-ownership

Network Database Model

- Member may have several owners



Network Database Model

- Advantages

- Conceptual simplicity, just like HDM
- Handles more relationship types (but all 1:M relationship)
- Data access flexibility
- Promotes database integrity
- Data independence
- Conformance to standards

Network Database Model

- Disadvantages

- System complexity
- Lack of structural independence

Relational Database Model (RDBM)

- Let's user or database designer to operate human logical environment
- Perceived by user as a collection of tables for data storage, while let RDBMS handles the physical details.
- Tables are a series of row/column intersections
- Tables related by sharing common entity characteristics
- It allows 1:1, 1:M, M:N relationships

Relational Database Model

Table name: AGENT

	AGENT_CODE	AGENT_LNAME	AGENT_FNAME	AGENT_INITIAL	AGENT_AREACODE	AGENT_PHONE
▶	501	Alby	Alex	B	713	228-1249
	502	Hahn	Leah	F	615	882-1244
	503	Okon	John	T	615	123-5589

Link through AGENT code

Table name: CUSTOMER

	CUS_CODE	CUS_LNAME	CUS_FNAME	CUS_INITIAL	CUS_AREACODE	CUS_PHONE	CUS_RENEW_DATE	AGENT_CODE
▶	10010	Ramas	Alfred	A	615	844-2573	05-Apr-2002	502
	10011	Dunne	Leona	K	713	894-1238	16-Jun-2002	501
	10012	Smith	Kathy	W	615	894-2285	29-Jan-2001	502
	10013	Olowski	Paul	F	615	894-2180	14-Oct-2002	502
	10014	Orlando	Myron		615	222-1672	28-Dec-2002	501
	10015	O'Brian	Amy	B	713	442-3381	22-Sep-2002	503
	10016	Brown	James	G	615	297-1228	25-Mar-2002	502
	10017	Williams	George		615	290-2556	17-Jul-2002	503
	10018	Farriss	Anne	G	713	382-7185	03-Dec-2002	501
	10019	Smith	Olette	K	615	297-3809	14-Mar-2002	503

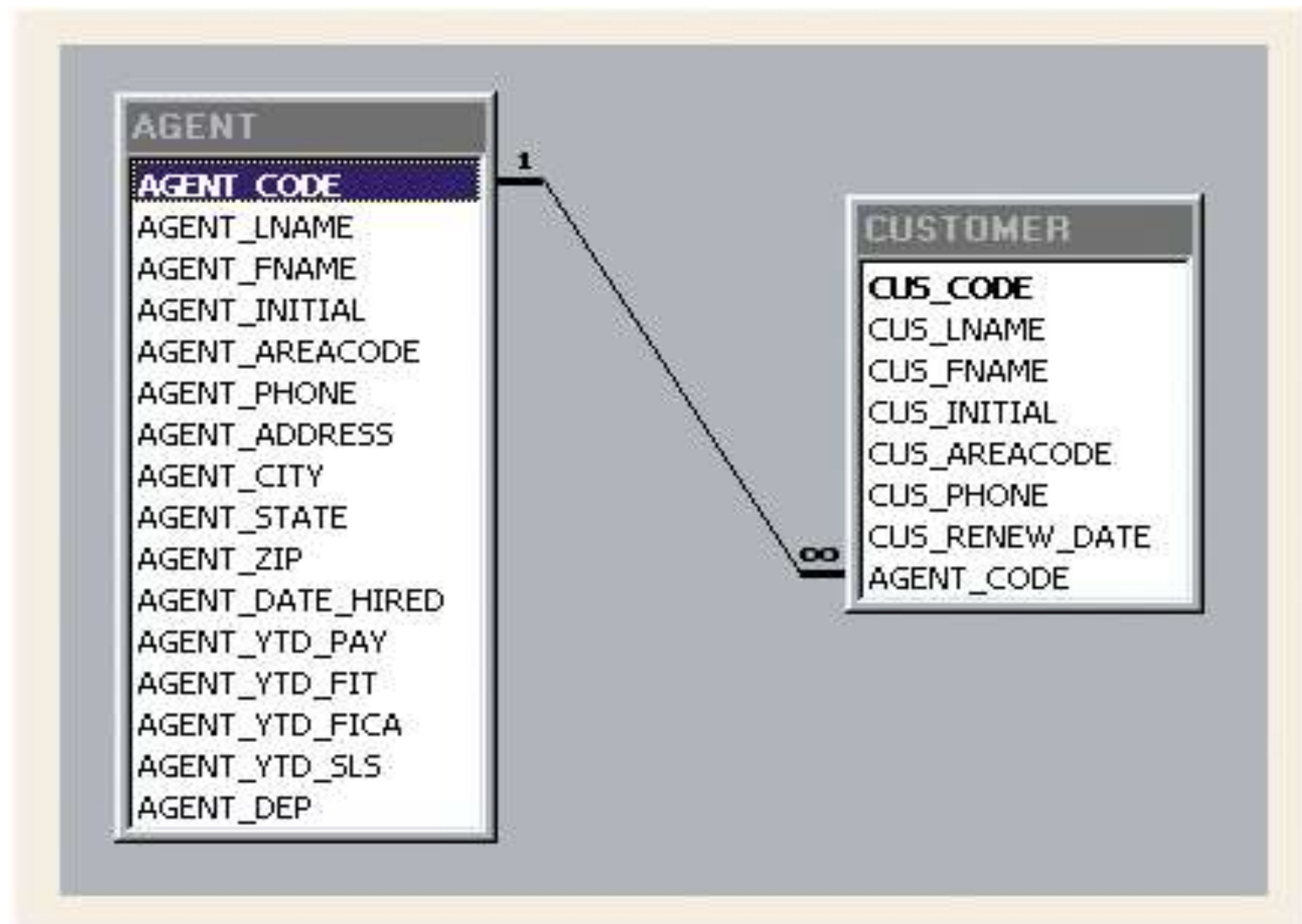


FIGURE 1.12 A RELATIONAL SCHEMA

Relational Database Model

- Advantages

- Ease of use
- Security
- Data Independence
- Data Manipulation Language
- Powerful database management system

Relational Database Model

- Disadvantages

- Substantial hardware and system software overhead
- Poor design and implementation is made easy
- May promote “islands of information” problems

- **Structured Limits**

- **Cost**

Entity Relationship Database Model (ERDBM)

- Complements the relational data model concepts
- ERDBM introduces a relational graphic representation
- ERDBM is based on several components
 - Entity, tabled entity (in RDM)
- Entity and entity set, a collection of like entities
- Each entity has attributes to describe the entity, which is similar to field in table
- Relationship and connection

Entity Relationship Database Model (ERDBM)

- Represented in an entity relationship diagram (ERD):
Chen's ERD model and Crow's Foot ERD
- Based on entities, attributes, and relationships

Entity Relationship Database Model

A one-to-many (1:M) relationship: a PAINTER can paint many PAINTINGs; each PAINTING is painted by one PAINTER



A many-to-many (M:N) relationship: an EMPLOYEE can learn many SKILLs; each SKILL can be learned by many EMPLOYEEs

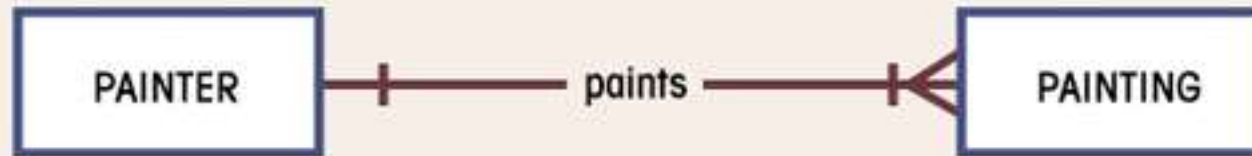


A one-to-one (1:1) relationship: an EMPLOYEE manages one STORE; each STORE is managed by one EMPLOYEE



FIGURE 1.13 RELATIONSHIP DEPICTION: THE ERD

A one-to-many (1:M) relationship: A PAINTER can paint many PAINTINGs;
each PAINTING is painted by one PAINTER



A many-to-many (M:N) relationship: an EMPLOYEE can learn many SKILLs;
each SKILL can be learned by many EMPLOYEEs



A one-to-one (1:1) relationship: an EMPLOYEE manages one STORE;
each STORE is managed by one EMPLOYEE



FIGURE 1.14 RELATIONSHIP DEPICTION: THE CROW'S FOOT ERD

Entity Relationship Database Model

- Advantages

- Exceptional conceptual simplicity
- Visual representation
- Effective communication tool
- Integrated with the relational database model

Entity Relationship Database Model

- Disadvantages

- No data manipulation language (no complete)

- Loss of information content

- Popular for high level design

Object-Oriented Database Model (OODBM)

- Semantic Data model (SDM)->Object-oriented Data Model (OODM)
- Object-oriented concept:
- Objects or abstractions of real-world entities are stored
 - Attributes describe properties
 - Collection of similar objects is a class, similar to entity set but contains procedure methods
 - Methods represent real world actions of classes
 - Classes are organized in a class hierarchy
 - Inheritance is the ability of object to inherit attributes and methods of classes above it

Object-Oriented Database Model (OODBM)

- Contains implementation and procedure operation information for more complicated data such as graphics, video, and other metadata
- Support transaction and information
- Reusability
- Portable to powered computing system

Object-Oriented Database Model

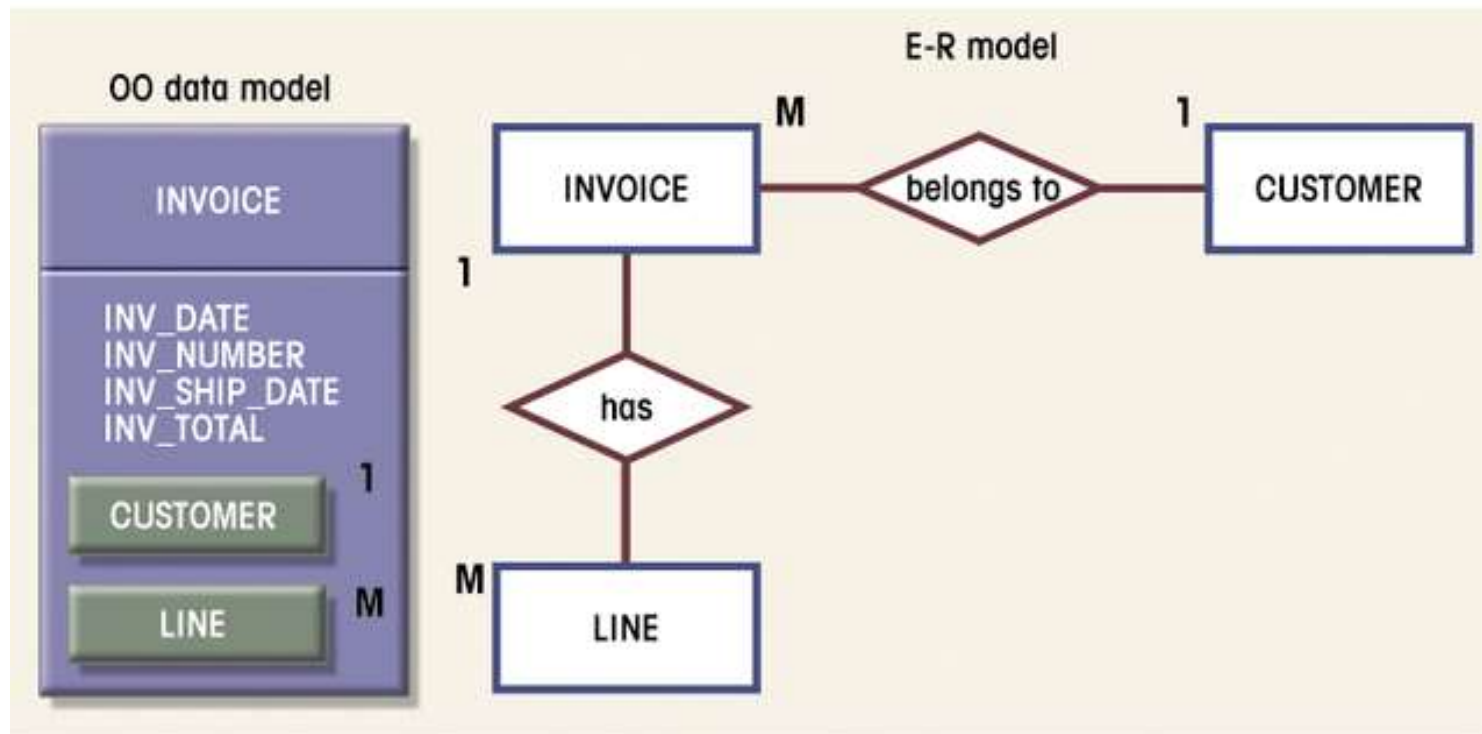


FIGURE 1.15 A COMPARISON OF THE OO DATA MODEL AND THE E-R MODEL

OO Database Model

- Advantages

- Adds semantic content
- Visual presentation includes semantic content
- Database integrity
- Both structural and data independence

OO Database Model

- Disadvantages
 - Lack of OODM
 - Complex navigational data access
 - Steep learning curve
 - High system overhead slows transactions

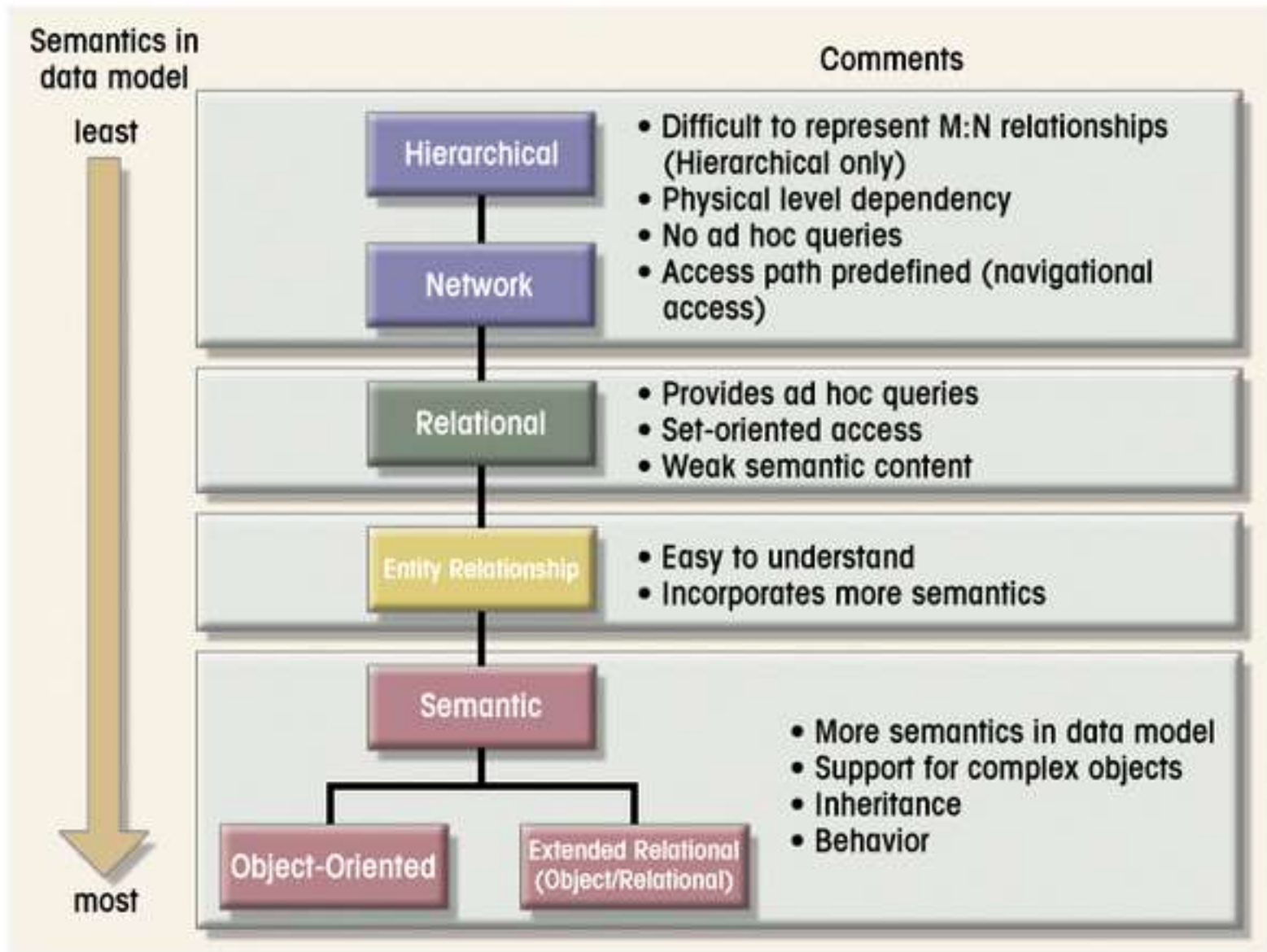


FIGURE 1.16 THE DEVELOPMENT OF DATA MODELS

Database Systems

- A database system is more than just data or data in combination with DBMS software
- A complete database system in an organization consists of four components:
 - Hardware
 - Software
 - Data and
 - People

Database Systems- Hardware

- The set of physical devices on which a database resides.
- The hardware portions of the system consist of:
 - The secondary storage volumes- typically magnetic disks- that are used to hold the stored data; together with the associated I/O devices, device controllers, I/O channels, and so forth.
 - The processor(s) and associated main memory that are used to support the execution of the database system software.

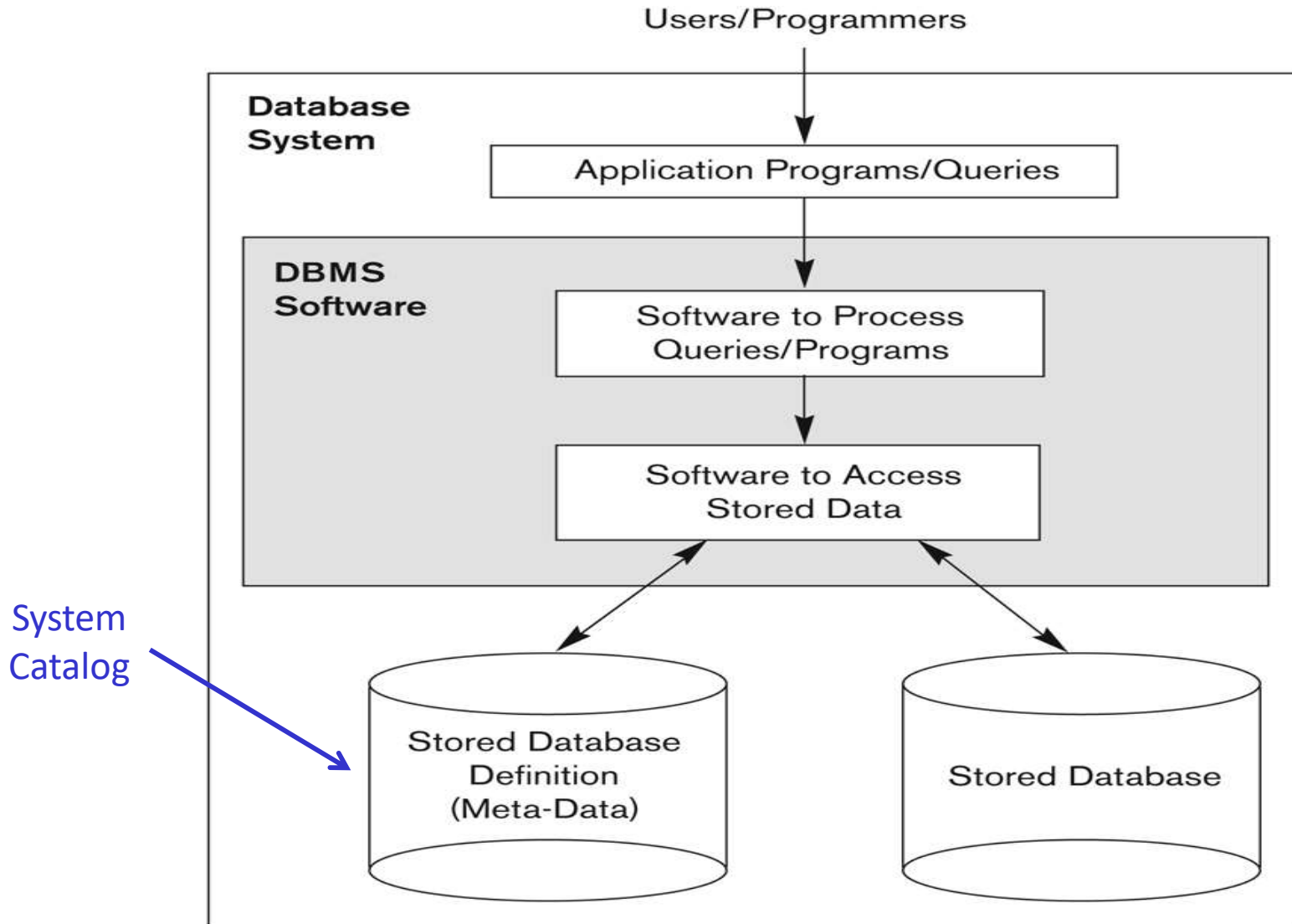
Database Systems- Software

- A database system includes two types of software
 - General-purpose database management software , usually called the Database Management System
 - Application software that uses DBMS facilities to manipulate the database to achieve a specific business function.

Database Management System

- A software system that enables users to **define, create, maintain, and control access to the database**
- A DBMS typically provides following services:
 - A centralized data definition & data control facility known as a data dictionary/directory (DD/D) or catalog
 - Data Security and integrity mechanisms
 - Concurrent data access for multiple users
 - User-oriented data query, manipulation, and reporting capabilities
 - Programmer-oriented application system development capabilities

A simplified database system environment



Database Systems- Data

- Used by the organization and a description of this data called the schema.
- In general, data in the database – at least in a large system – will be both ***integrated*** and ***shared***.
- Integrated means that the database can be thought of as a unification of several otherwise distinct data files, with any redundancy among those files either wholly or partly eliminated.
- Shared means that individual piece of data in the database can be shared among several different users, in the sense that each of those users can have access to the same piece of data.

Database Systems - People

- Data Administrator (DA), Database Administrator (DBA), Database Designers (Logical and Physical), Application Programmers, End Users (naive and sophisticated)

References

- Connolly, T. M. & Begg, C. E. Database Systems A Practical Approach to Design, Implementation, and Management.
- Gray W. Hansen, James V. Hansen ,Database Management and Design.
- Ramez Elmasri and Shamkant B. Navathe, Fundamentals of Database Systems